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I have a great interest in Ai/ML and Statistics but it was a good experience of making web app from scratch for the very first time.

Description

The Quiz Master app includes 2 roles-admin and user, wherein the admin manages subjects,chapters,quizzes and questions while the user can attempt quizzes of different subjects multiple times and keep record of their scores.

Technologies used

- Flask - for Backend
- Flask-SQLAlchemy - ORM for database management
- SQLite - Database for storing quiz data
- HTML,CSS - For Frontend and Styling
- Jinja2 - for templating

DB Schema Design

Database Tables and Columns:

1. Users(id,email,password,full_name,role)
2. Subjects(id,name)
3. Chapters(id,name,subject_id)
4. Quizzes(id,title,chapter_id,date,duration)
5. Questions(id,text,option_a,option_b,option_c,option_d,correct_option,quiz_id)
6. Scores(id,user_id,quiz_id,score)

[Database Schema](#)

API Design

Not Implemented

Architecture and Features

Project Structure:

app.py - Main flask app file. It also handles all the endpoints for admin and user functionalities

models.py - Contains database schema using Flask-SQLAlchemy

templates/ - Contains all the html files for the frontend

static/ - Contains style.css for styling

Features Implemented:

- Admin Features
 - Manage Users,Subjects,Chapters,Quizzes,Questions.
 - Set Quiz Date and Duration.
 - Can view performance of students in the quizzes.
- User Features:
 - Take the quizzes multiple times.
 - View current and past scores.

Video

 Project Demonstration.mp4